

DINESH CHAHAR

IIT Delhi | Backend Software Engineer — Java, Spring Boot, Microservice & Distributed Systems

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WORK EXPERIENCE

Flixstock

Software Developer

Jun '25 – Feb '26

Gurugram, India

Distributed Task Execution Service | Python, FastApi, Ray, RabbitMQ, Redis

- Architected a **Ray-based distributed** task execution service with a **Redis-backed, resource-aware scheduler** across cluster nodes.
- Improved ML batch throughput **35%** via **idempotent execution**, guaranteeing exactly-once effect despite at-least-once delivery.
- Optimized scheduling with **RabbitMQ direct-exchange priority queues**, reducing p99 task latency **28%** for critical workloads.
- Engineered fault tolerance via worker heartbeats, a **reaper** reclaiming dead-worker tasks, and **circuit-breaker**-guarded **DLQ** retries.

AI Media Workflow Orchestration Platform | Java, Spring Boot, Spring Security, lambda, s3, SQS, PostgreSQL

- Built a Spring Boot service orchestrating **QC-editor reviews** for **30K+ SKUs daily**, then a **Lambda** handoff to Client Service.
- Modeled each SKU's review as a **bounded-retry state machine**, capping revisions at **3 cycles** before a terminal **Not Deliverable** status.
- Enforced QC/editor role separation via **Spring Security RBAC** and **stateless JWT authentication** across the review pipeline.
- Persisted every editor revision and QC remark as **immutable, append-only history**, enabling full **audit trails** and version rollback.

Tipping Mr. Pink (Burger Singh)

Software Developer

Jul '24 – Apr '25

Gurugram, India

- Built a geospatial intelligence platform for store expansion, reducing location analysis latency by **22%** using proximity caching.
- Implemented **Redis GEO** indexing and **GeoHash** based deduplication, reducing redundant **Google APIs** requests and external costs.
- Developed asynchronous job pipeline using **RabbitMQ** workers to process external API calls without blocking request threads.
- Designed scalable **SpringBoot** backend with **PostGIS** spatial queries and domain-restricted **OAuth** for secure internal analytics.

Prototech Solutions

Software Development Intern

May '22 – Jul '22

Pune, India

- Automated 3D drafting plugins using **C# and .NET**, reducing manual design cycles by **40%** for civil engineering clients.
- Enhanced accuracy by **10%** by implementing complex geometric algorithms directly within the **Advance Steel API** layer.
- Conducted comprehensive unit and integration testing to ensure stability across critical distributed software design modules.

TECHNICAL PROJECTS

Distributed Cache | ConcurrentHashMap, Java NIO, Custom TCP Protocol, JMH

- Built a **Redis-like distributed cache** in Java implementing **LRU** eviction, **TTL** expiration, and scheduled background cleanup threads
- Designed **lock-striped** keyspace segmentation with **32 partitions** via **ReentrantReadWriteLock** allowing parallel non-blocking reads.
- Built custom 200-vnode consistent hashing using TreeMap for uniform key distribution during **dynamic** cluster membership changes.
- Developed 12-byte binary protocol over NIO vs HTTP that consistently reached **0.8ms** p99 under 10K concurrent operations via JMH.

Real-Time Inventory Reservation & Dark Store Router | Java, Spring Boot, Redis Lua, Kafka, PostgreSQL, WebSocket

- Engineered checkout logic selecting dark stores by scoring proximity, inventory availability, and real-time store workload metrics
- Implemented atomic cart reservations via **Redis Lua scripts**, validated by **high-concurrency** testing with fifty threads for accuracy.
- Synchronized Redis state to Postgres via Kafka using **manual offsets** and **idempotent** consumers preventing stock double-counting.
- Integrated **WebSocket** live stock updates and POS communication over raw TCP sockets using **16-byte** binary frame formats.

EDUCATION

Indian Institute of Technology (IIT), Delhi

Bachelor of Technology (B.Tech) in Mechanical Engineering

2019 – 2024

New Delhi, India

TECHNICAL SKILLS

Languages: Java, Python, SQL, C#

Frameworks & Backend: Spring Boot, FastAPI, Django, .NET Core, RESTful APIs, Asynchronous Processing, Idempotency

Data, Messaging & Infrastructure: Kafka, RabbitMQ, Redis, Ray, Celery, Postgres, MySQL, AWS (EC2, S3), Docker, NGINX, Git, CI/CD, Transactional Outbox, High Concurrency